

QMEA (Problem-Solving Questions)

1. How does an MBA major affect the number of job offers received? An MBA student randomly sampled four recent graduates, one each in finance, marketing, and management, and asked them to report the number of job offers. Use a 5% significance level.

Finance	Marketing	Management
3	1	8
1	5	5
4	3	4
1	4	6

The following ANOVA table was obtained:

Source of Variation	SS	df	MS	F	p-value
Between Groups (Treatments)	26.00	2	13.00	4.825	0.038
Within Groups (Error)	24.25	9	2.694		
Total	50.25	11			

Questions

- State the null hypothesis and alternative hypothesis.
- Based on the ANOVA table, should H_0 be rejected or not rejected?
- What can you conclude about the mean number of job offers among the three MBA majors?

2. A consumer organization was concerned about the differences between the advertised sizes of containers and the actual amount of product. In a preliminary study, six packages of three different brands of margarine that are supposed to contain 500 ml were measured. The differences from 500 ml are listed here. Use a 1% significance level.

Brand 1	Brand 2	Brand 3
1	2	1
3	2	2
3	4	4
0	3	2
1	0	3
0	4	4

The following ANOVA results were obtained:

Source of Variation	SS	df	MS	F	p-value
Between Groups (Treatments)	6.333	2	3.167		0.218
Within Groups (Error)	28.167	15	1.878		
Total	34.500	17			

Questions

- State the null hypothesis and alternative hypothesis.
- Find F-statistic
- Based on the ANOVA table, should H_0 be rejected or not rejected?
- What can you conclude about the mean differences from 500 ml among the three brands?

3. As an experiment to understand measurement error, a statistics professor asks four students to measure the height of the professor, a male student, and a female student. The differences (in centimeters) between the correct dimension and the ones produced by the students are listed here.

Student	Professor	Male Student	Female Student
1	1.4	1.5	1.3
2	3.1	2.6	2.4
3	2.8	2.1	1.5
4	3.4	3.6	2.9

The following Randomized Block ANOVA results were obtained:

Source of Variation	SS	df	MS	F	p-value
Treatments	0.872	2	0.436	5.078	0.051
Blocks	5.910	3	1.970	—	0.001
Error	0.515	6	0.086		
Total	7.297	11			

Use a 5% significance level.

- Identify the treatments and the blocks in this experiment.
- State the null hypothesis and alternative hypothesis.
- Find F-statistic
- Based on the ANOVA table, should H_0 be rejected or not rejected?
- What can you conclude about the mean measurement errors among the three subjects?

4. A university surveyed 150 students about their preferred mode of transportation to campus. Based on previous studies, the expected proportions for the five transportation modes are 10%, 20%, 30%, 20%, and 20%, respectively.

The observed results are shown below.

Transportation Mode	Expected Proportion	Observed Frequency
Bus	0.10	12
Car	0.20	32
Motorcycle	0.30	42
Bicycle	0.20	36
Walking	0.20	28

The following statistical result was obtained:

Test	Test Statistic	df	p-value
_____	2.267	4	0.687

Using a 5% significance level, answer the following questions:

- Identify the appropriate statistical test.
- State the null hypothesis and alternative hypothesis.
- Based on the table, should H_0 be rejected or not rejected?
- State your conclusion.

5. The operations manager of a company that manufactures shirts wants to determine whether there are differences in the quality of workmanship among the three daily shifts. She randomly selects 600 recently made shirts. Each shirt is classified as either perfect or flawed, and the shift that produced it is recorded.

The observed results are shown below.

Shirt Condition	Shift 1	Shift 2	Shift 3
Perfect	240	191	139
Flawed	10	9	11

The following statistical result was obtained:

Statistical Test	Test Statistic	df	p-value
_____	2.35	2	0.309

Using a 5% significance level, answer the following questions:

- Identify the appropriate statistical test.
- State the null and alternative hypotheses.
- Based on the result above, should the null hypothesis be rejected?
- State your conclusion in the context of the problem.

6. The marketing manager for a chain of hardware stores needed more information about the effectiveness of the three types of advertising that the chain used. These are localized direct mailing (in which flyers describing sales and featured products are distributed to homes in the area surrounding a store), newspaper advertising, and local television advertisements. To determine which type is most effective, the manager collected 1 week's data from 100 randomly selected stores. For each store, the following variables were recorded: Weekly gross sales Weekly expenditures on direct mailing Weekly expenditures on newspaper advertising Weekly expenditures on television commercials All variables were recorded in thousands of dollars. Use a 5% significance level

- Find the regression equation.
- What are the coefficient of determination and the coefficient of determination adjusted for degrees of freedom? What do these statistics tell you about the regression equation?
- What does the standard error of estimate tell you about the regression model?
- Test the validity of the model.
- Which independent variables are linearly related to weekly gross sales in this model? Explain.
- Compute the 95% interval of the week's gross sales if a local store spent \$800 on direct mailing, \$1,200 on newspaper advertisements, and \$2,000 on television commercials.
- Calculate the 95% interval of the mean weekly gross sales for all stores that spend \$800 on direct mailing, \$1,200 on newspaper advertising, and \$2,000 on television commercials.
- Discuss the difference between the two intervals found in parts (f) and (g)

	A	B	C	D	E	F
1	SUMMARY OUTPUT					
2						
3	<i>Regression Statistics</i>					
4	Multiple R	0.4419				
5	R Square	0.1953				
6	Adjusted R Square	0.0803				
7	Standard Error	2.59				
8	Observations	25				
9						
10	<i>ANOVA</i>					
11		<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
12	Regression	3	34.10	11.37	1.70	0.1979
13	Residual	21	140.56	6.69		
14	Total	24	174.66			
15						
16		<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	
17	Intercept	12.31	4.70	2.62	0.0160	
18	Direct	0.570	1.72	0.331	0.7437	
19	Newspaper	3.32	1.54	2.16	0.0427	
20	Television	0.732	1.96	0.374	0.7123	

	A	B
1	Prediction Interval	
2		
3		Sales
4		
5	Predicted value	18.21
6		
7	Prediction Interval	
8	Lower limit	12.27
9	Upper limit	24.15
10		
11	Interval Estimate of Expected Value	
12	Lower limit	15.70
13	Upper limit	20.73

7. For many cities around the world, garbage is an increasing problem. Many North American cities have virtually run out of space to dump the garbage. A consultant for a large American city decided to gather data about the problem. She took a random sample of houses and determined the following:

Y = the amount of garbage per average week (pounds)

X1 = Size of the house (square feet)

X2 = Number of children

X3 = Number of adults who are usually home during the day

a. Conduct a regression analysis.

b. Is the model valid? Use a 5% significance level

c. Interpret each of the coefficients.

d. Test to determine whether each of the independent variables is linearly related to the dependent variable.

	A	B	C	D	E	F
1	SUMMARY OUTPUT					
2						
3	<i>Regression Statistics</i>					
4	Multiple R	0.4125				
5	R Square	0.1702				
6	Adjusted R Square	0.1645				
7	Standard Error	3.67				
8	Observations	440				
9						
10	ANOVA					
11		<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
12	Regression	3	1205.9	401.98	29.80	1.52E-17
13	Residual	436	5880.4	13.49		
14	Total	439	7086.3			
15						
16		<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	
17	Intercept	7.19	1.09	6.59	1.30E-10	
18	House Size	0.00190	0.00059	3.21	0.0014	
19	Children	1.10	0.141	7.84	3.58E-14	
20	Adults	1.04	0.233	4.48	9.58E-06	

8) Lotteries have become important sources of revenue for governments. Many people have criticized lotteries, however, referring to them as a tax on the poor and uneducated. In an examination of the issue, a random sample of 100 adults was asked how much they spend on lottery tickets and was interviewed about various socioeconomic variables. Use a 5% significance level

The purpose of this study is to test the following beliefs:

1. Relatively uneducated people spend more on lotteries than do relatively educated people.
2. Older people buy more lottery tickets than younger people.
3. People with more children spend more on lotteries than people with fewer children.
4. Relatively poor people spend a greater proportion of their income on lotteries than relatively rich people.

The following data were recorded: Amount spent on lottery tickets as a percentage of total household income, Number of years of education, Age, Number of children, Personal income (in thousands of dollars)

- a. Develop the multiple regression equation.
- b. Is the model valid?
- c. Test each of the beliefs. What conclusions can you draw?

	A	B	C	D	E	F
1	SUMMARY OUTPUT					
2						
3	<i>Regression Statistics</i>					
4	Multiple R	0.6584				
5	R Square	0.4335				
6	Adjusted R Square	0.4096				
7	Standard Error	2.91				
8	Observations	100				
9						
10	ANOVA					
11		<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
12	Regression	4	615.44	153.86	18.17	4.09E-11
13	Residual	95	804.35	8.47		
14	Total	99	1419.79			
15						
16		<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	
17	Intercept	11.91	1.79	6.67	1.69E-09	
18	Education	-0.430	0.132	-3.26	0.0016	
19	Age	0.0292	0.0252	1.16	0.2501	
20	Children	0.0934	0.224	0.417	0.6780	
21	Income	-0.0745	0.0277	-2.69	0.0085	